

Rank&File

Smart Resume Screening & Candidate Ranking

AI-Powered Resume Tailoring · Neural Vector Matching · Structural ATS
Optimization

NODE.JS

EXPRESS

PYTHON FASTAPI

ARTIFICIAL INTELLIGENCE



Challenges in Modern Recruitment

Resume Overload

Hiring managers receive hundreds of applications per role, making manual review slow, biased, and error-prone.

The ATS "Black Box"

Job seekers are rejected by Applicant Tracking Systems due to formatting issues or missing technical keywords — often without knowing why.

Shallow Keyword Matchers

Traditional tools use simple string matching, failing to recognize contextual relationships — e.g., not connecting **PyTorch** to **Deep Learning**.



A Dual-Target Hybrid AI Platform

Rank&File serves both sides of the hiring equation with a single intelligent engine.



For Job Seekers

Instantly analyzes resumes against job postings, identifies skill gaps, and generates a 1-click ATS-tailored resume.



For Recruiters

Automatically screens candidate pools, ranks applicants by composite match score, and exports ranked data to CSV.



Hybrid AI Engine

Combines fast client-side JS matching with deep Python neural embeddings and real-time Groq LLM assistance.

Platform Capabilities



Decomposed Sub-Scores

Displays Hard Skill %, TF-IDF Vector Similarity %, and ATS Audit % separately for full transparency.



Domain Skill Taxonomy

Auto-extracts skills across Frontend, Backend, Cloud/DevOps, AI/Data, Mobile, Cybersecurity, and more.



Multi-Format Parsing

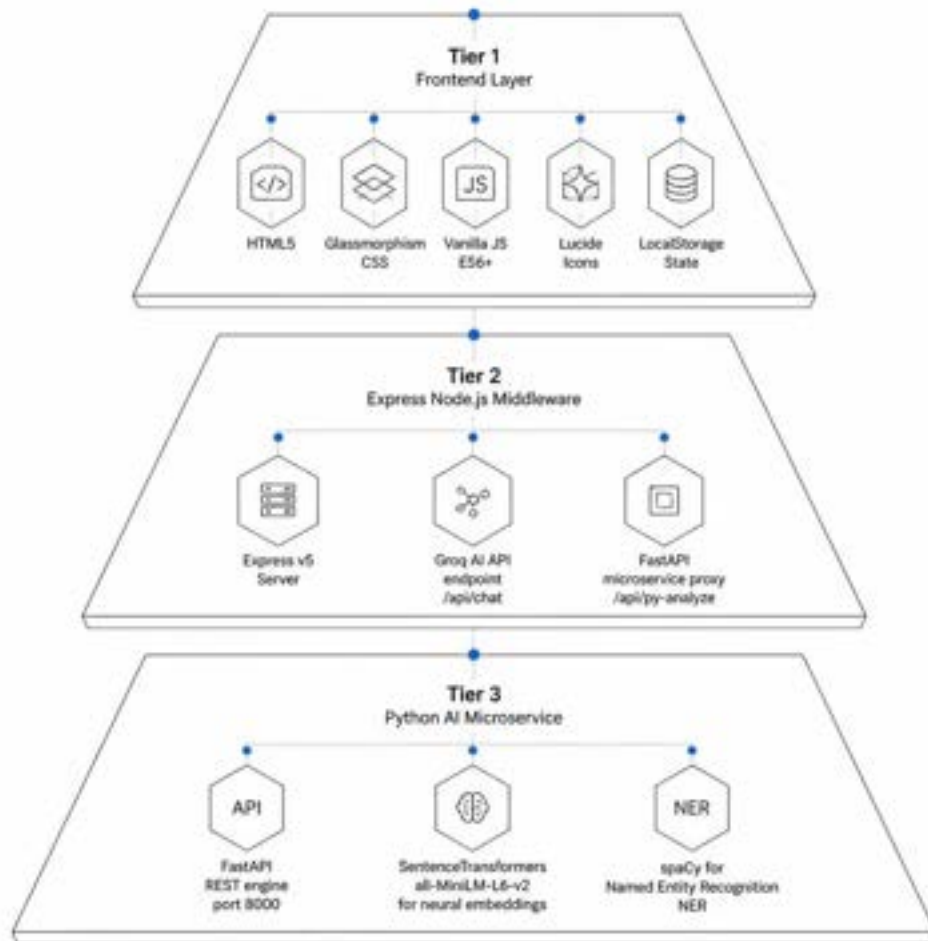
Native handling of .TXT and .PDF resumes using pdf.js for seamless ingestion.



AI Chatbot

Interactive assistant powered by llama-3.3-70b-versatile for real-time resume guidance and interview prep questions.

End-to-End System Architecture



A Clean 3-Tier Stack

Rank&File separates concerns across three distinct layers for maintainability and scalability.

- **Frontend:** Glassmorphism UI with Vanilla JS and LocalStorage state management
- **Middleware:** Express v5 proxies AI requests and serves static assets
- **AI Engine:** FastAPI microservice running SentenceTransformers and spaCy NER

Multi-Metric Scoring Model

Composite Match Score Formula

$$\text{Score} = (0.50 \times \text{Hard Skills}) + (0.30 \times \text{Semantic Vector}) + (0.20 \times \text{ATS Structural})$$

50% — Hard Skills

Direct match ratio of required technologies in candidate text vs. job requirements.

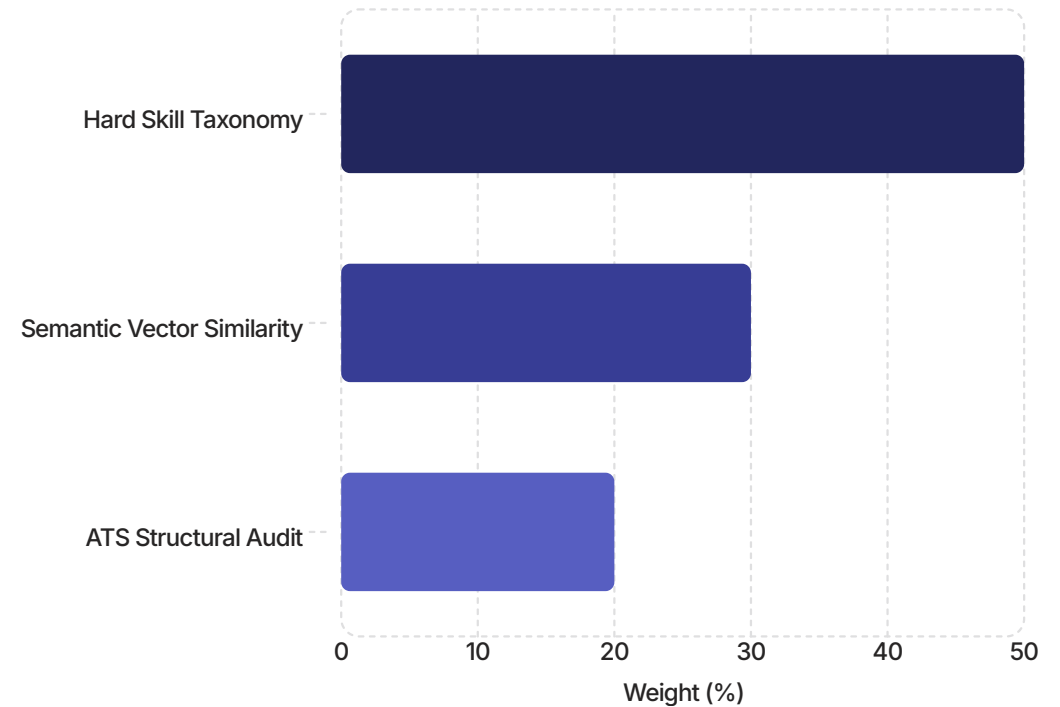
30% — Semantic Vector

Neural cosine similarity using TF-IDF token vectors and SentenceTransformer embeddings.

20% — ATS Structural

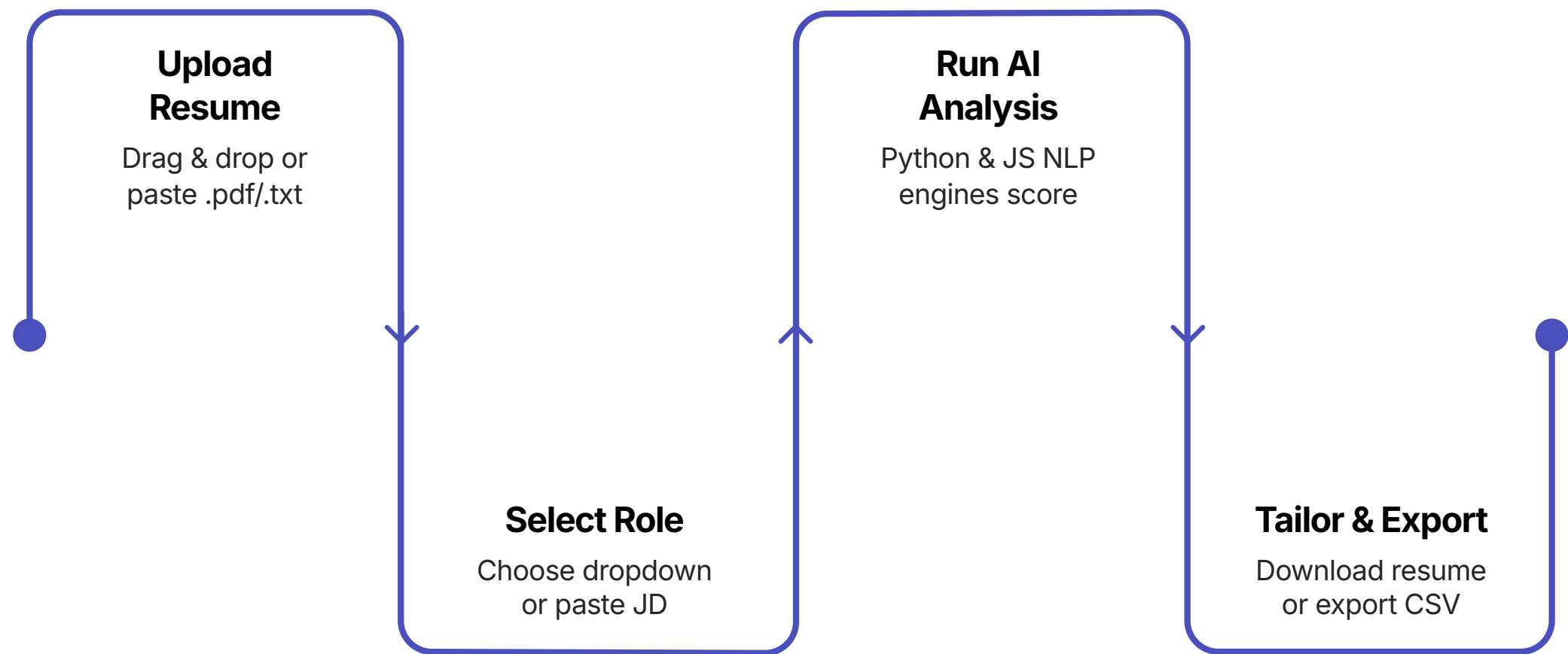
Mechanical check for contact info, headings, action verbs, and quantified impact metrics.

Component



How To Use Rank&File

A streamlined four-step journey for both candidates and recruiters.



Each step is designed for zero-friction interaction — from resume upload to ranked candidate export in under 60 seconds.

Local Deployment & Execution Guide

Terminal 1 — Python AI Engine (Port 8000)

```
pip install -r requirements.txt
python -m spacy download
en_core_web_sm
python main.py
```

Terminal 2 — Express Web Server (Port 3000)

```
npm install
npm run dev
```

Terminal 3 — Live Access

Open your browser to
<http://localhost:3000>



📄 The application runs fully offline with graceful fallbacks if external microservices are unreachable.

Live Demo Checklist

01

Skill Analyzer

Upload a sample resume, select **AI & Data Scientist**, and run analysis to demonstrate multi-gauge meters.

02

Actionable Outputs

Point out missing skills, ATS structural suggestions, and auto-generated technical interview questions.

03

1-Click Tailored Resume

Show the generated plain-text output and use the **Copy Text / Download (.TXT)** controls.

04

Ranking Portal

Switch tabs, select a position, demonstrate automated candidate sorting by rank, and click **Export CSV**.

05

AI Chatbot

Open the floating widget and ask: *"How can I improve my ATS score?"*



Summary & Project Impact

70%

Time Saved

Automates candidate screening workflows for recruiters, drastically reducing manual review time.

Multiple

Skill Domains

Comprehensive taxonomy covering every major technology and professional competency area.

3

Score Metrics

Hard Skills, Semantic Vector, and ATS Structural — a complete, transparent evaluation model.

- ✓ Rank&File delivers higher interview callbacks for job seekers and smarter, faster hiring decisions for recruiters — all powered by a resilient, locally-runnable architecture.

Thank you!